POLITECNICO MILANO 1863

BASI DI DATI 1 – Anno Accademico 2024-2025

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Part 1 – ON A SEPARATE SHEET FROM PART 2

Exercise 1 – SQL (13pt)

Consider the following database of transactions between banks in various countries, where the creditor is the bank that lends money while the debtor is the bank that borrows it and must return the amount (stored, therefore, as a positive value) after a specified number of months. For some transactions, the duration may not be specified. Each bank may only operate in one sector ('insurance', 'credit', etc.) and in one country. We also want to keep track of the supervisory bodies that monitor banks on specific sectors. Note that some institutions supervise banks in more than one country. In some countries, the same institution may deal with several sectors, and one sector may be monitored by several institutions.

TRANSACTION(ID, Creditor, Debtor, TransactionDate, TransactionTime, Amount, Duration)

BANK(RegisterNumber, Sector, CountryOfRegistration, Name)

SUPERVISION(BodyName, SupervisedSector, SupervisedCountry, Scope)

A. Specify in SQL the creation of the TRANSACTION table, defining all necessary constraints. (2 pt)
B. Find, for each day, the average amount of transactions whose duration is above one month, where both counterparties are supervised by at least one 'supranational' body. (4 pt)
C. For each country, find the banks supervised by the smallest number of institutions. Note: we are also interested in banks without supervisory bodies, a situation that is possible in some exotic countries. (4 pt)
D.

Exercise 2 – Formal languages (6pt)

Write the following query in Relational Algebra (3pt) and Tuple Relational Calculus (3pt): find Italian banks that have only carried out transactions with a duration of more than 24 months, either as creditors or as debtors.

PART 2 – ON A SEPARATE SHEET FROM PART 1

Exercise 3 – Theory question (1pt): Explain the relationship between minimum cardinality of a relation in the ER model and the 'NOT NULL' constraint in the relational model.

Exercise 4 – Conceptual and logical design (13pt)

- *The project evaluation is based on its completeness, correctness, legibility, minimality and clarity. Remember to specify the **identifiers** of all the entities and the **cardinality** of all the associations.*
- *Disporre il progetto concettuale e il progetto logico nelle pagine interne del foglio; arrange the conceptual project and the logical project on the inside pages of the sheet; the logical project is also a graph, with nodes for tables and arcs for referential integrity, which must reflect the position of entities and relations in the conceptual project.*

A delivery startup offers a service that delivers restaurant orders to end customers through a digital platform and a network of couriers. Each registered customer has a personal profile that includes their first name, last name, email address, phone number, and one or more associated delivery addresses. When customers place orders, they select one of their delivery addresses. Each order has a unique identifier, a registration date and time, a total amount paid by the customer, and a status (e.g., "In Preparation," "Delivered"). An order may include items from one or more restaurants.

For each restaurant, the database records name, address, cuisine type, and phone number. Restaurants can also indicate weekly opening and closing days and hours, if available. Each restaurant manages its menu, which consists of various items (dishes/beverages). Each item is identified by a unique code within the restaurant and has a name, description, price, and availability status.

For each courier, personal data, phone number, vehicle type, and employment category (employee or freelancer) are recorded. Employee couriers receive a fixed salary, while freelancers are paid based on completed deliveries and can sign multiple contracts with the startup. Each contract includes details such as a start date, an end date, an agreed delivery fee, and contractual terms.

Orders are delivered to customers by a single courier. Operational details such as the actual delivery time and any issues encountered are recorded for each delivery. To improve service quality, the startup constantly monitors courier performance by collecting data on the number of completed deliveries and customer feedback.

Each restaurant can offer percentage-based discounts to promote specific items for limited periods. For example, a restaurant may offer a 10% discount on Margherita pizza for a week or a month. If a promotion is particularly popular, the restaurant may repeat it.